

# Spring 2013

**Problem 1 (Sp13).** Suppose  $f : \mathbb{R} \rightarrow \mathbb{R}$  is a bounded continuous function. Calculate the limit

$$\lim_{\varepsilon \rightarrow 0^+} \int_{-\infty}^{\infty} f(t) \frac{\varepsilon}{\varepsilon^2 + t^2} dt$$

**Problem 2 (Sp13).** Suppose that  $f$  is a smooth real function defined for all real  $x$ , such that  $|f'(x)| \geq \varepsilon > 0$  and  $|f''(x)| \leq M > 0$  for all  $x$ .

1. Show that  $f$  has a unique zero  $z$ .
2. Given  $x_0$ , define a sequence by  $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ . Show that

$$|x_{n+1} - z| \leq |x_n - z|^2 \frac{M}{\varepsilon}$$

3. Show that the sequence  $(x_n)$  converges to the zero  $z$  of  $f$  provided that  $|f(x_0)| < \varepsilon^2/M$ .

**Problem 3 (Sp13).** Show that  $\int_0^{\infty} x \exp(-x^6 \sin^2 x) dx$  is finite.

**Problem 4 (Fa92, Sp13).** Evaluate  $\int_C \frac{e^z}{z(2z+1)^2} dz$  where  $C$  is the unit circle with counterclockwise orientation.

**Problem 5 (Sp13).** Let  $n \geq 1$  and let  $\{a_0, a_1, \dots, a_n\}$  be complex numbers such that  $a_n \neq 0$ . For  $\theta \in \mathbb{R}$ , define

$$f(\theta) = a_0 + a_1 e^{i\theta} + a_2 e^{2i\theta} + \dots + a_n e^{ni\theta}$$

Prove that there exists  $\theta \in \mathbb{R}$  such that  $|f(\theta)| > |a_0|$ .

**Problem 6 (Sp13).** Show that if  $V$  is a real vector space with a positive definite symmetric bilinear form and  $W \subset V$  is a linear subspace then  $W^\perp = ((W^\perp)^\perp)^\perp$ . Give an example such that  $W \neq (W^\perp)^\perp$ .

**Problem 7 (Sp90, Fa93, Sp13, Sp18, Sp19, Sp22).** 1. If  $A$  is a matrix over an algebraically closed field of characteristic 0 with  $A^n = I$  for some  $n > 0$  show that  $A$  is diagonalizable.

2. Show that over any field of characteristic  $p > 0$  there is a matrix  $A$ , with  $A^n = I$  for some  $n > 0$ , that is not diagonalizable.

**Problem 8 (Sp13).** Let  $m$  and  $n$  be integers greater than 1. Prove that  $\log_m n$  is rational if and only if  $m = l^r$  and  $n = l^s$ , for some positive integers  $l, r$ , and  $s$ .

**Problem 9 (Sp13).** Let  $\mathbb{K}$  be a field. Let  $R$  be an integral domain which contains  $\mathbb{K}$  and is finite dimensional (as a vector space) over  $\mathbb{K}$ . Prove that  $R$  is a field.

**Problem 10 (Sp13).** Find  $\int_0^1 \arctan x \, dx$ .

**Problem 11 (Sp06, Fa10, Sp13, Sp21).** Prove that the intersection of a decreasing sequence of nonempty closed connected subsets of a compact metric space is nonempty and connected. Give an example to show that this is false if the assumption that the space is compact is dropped.

**Problem 12 (Fa80, Sp13).** Let  $g$  be  $2\pi$ -periodic, continuous on  $[-\pi, \pi]$  and have Fourier series

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Let  $f$  be  $2\pi$ -periodic and satisfy the differential equation

$$f''(x) + kf(x) = g(x)$$

where  $k \neq n^2, n = 1, 2, 3, \dots$ . Find the Fourier series of  $f$  and prove that it converges everywhere.

**Problem 13 (Sp13).** Let  $U$  be an open subset of  $\mathbb{C}$ . Let  $K$  be a closed bounded subset of  $\mathbb{C}$  that is contained in  $U$ . Put

$$D = \min_{p \in K, q \notin U} |p - q|$$

That is,  $D$  is the closest distance between  $K$  and  $\mathbb{C} - U$ . (If  $U = \mathbb{C}$  then we put  $D = \infty$ .) Suppose that  $f$  is an analytic function on  $U$  such that for all  $z \in U$ , we have  $|f(z)| \leq M$ . Find an explicit number  $C < \infty$ , depending on  $M$  and  $D$ , such that for all  $z_0 \in K$  we have  $|f'(z_0)| \leq C$ .

**Problem 14 (Sp13).** Which of the following domains are biholomorphically equivalent to each other: the complex plane  $\mathbb{C}$ , the unit disc  $\mathbb{D}$  and the upper half-plane  $\mathbb{H}$ ?

**Problem 15 (Sp13).** Show that the  $n \times n$  Cauchy matrix with entries  $a_{ij} = 1/(x_i - y_j)$ , where  $x_i, y_j$  are  $2n$  different numbers, has determinant

$$\frac{\prod_{1 \leq j < i \leq n} (x_i - x_j) (y_j - y_i)}{\prod_{1 \leq i, j \leq n} (x_i - y_j)}$$

**Problem 16 (Fa81, Sp13, Fa22).** Prove the following three statements about real  $n \times n$  matrices.

1. If  $A$  is an orthogonal matrix whose eigenvalues are all different from  $-1$ , then  $I + A$  is nonsingular and  $S = (I - A)(I + A)^{-1}$  is skew-symmetric.
2. If  $S$  is a skew-symmetric matrix, then  $A = (I - S)(I + S)^{-1}$  is an orthogonal matrix with no eigenvalue equal to  $-1$ .
3. The correspondence  $A \leftrightarrow S$ , called the Cayley transform, from Parts 1 and 2 is one-to-one.

**Problem 17 (Sp13).** Consider the symmetric group  $S_n$  in its presentation as  $n \times n$  permutation matrices. Define the *expected trace* to be the weighted sum of traces

$$E_n = \frac{1}{n!} \sum_{g \in S_n} \text{tr } g$$

Calculate  $E_n$ .

**Problem 18 (Sp13).** Show that more than half the elements of a finite field  $\mathbb{F}$  are squares. Show that every element is the sum of 2 squares.